POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6011T | Semester VI | Theory | 4 Credits

Transportation Engineering

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6011T as Transportation Engineering in Semester VI for Civil Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kharagpur’s Introduction to Transportation Engineering course, NPTEL IIT Roorkee’s Transportation Engineering II course and the MSBTE polytechnic syllabus. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Highway alignments, pavement designs, bridge/tunnel structures and airport layouts must be based on approved engineering designs, current IRC/railway/aviation codes, authoritative site data and competent professional review.

Verified source note: Verified sources support highway alignment/geometric design, traffic studies, pavement design (IRC methods), railway engineering (gauges, permanent way, signals), airport engineering (site selection, runways) and bridge/tunnel basics. The four-module sequence is a learning sequence, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Transportation Engineering studies the planning, design, construction and maintenance of facilities for the safe and efficient movement of people and goods. It develops the ability to design highway alignments, analyze pavement structures, understand railway track systems, coordinate traffic studies, and plan airport, bridge and tunnel infrastructure while respecting the safety and regulatory boundaries of transportation systems.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Basic surveying, engineering mechanics, soil mechanics, hydraulics, engineering graphics and measurement awareness.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain transportation fundamentals, highway alignment, geometric design and traffic stream characteristics.
2. Apply principles of pavement design and maintenance for flexible and rigid road structures.
3. Explain railway engineering concepts, track components, geometric design and signaling systems.
4. Explain the planning and components of bridges, tunnels and airports, including site selection and visual aids.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain highway planning, alignment factors and the geometric design of road elements.	Understand
CO2	Apply pavement design principles and maintenance strategies for sustainable road infrastructure.	Apply
CO3	Explain railway track components, geometric design of tracks and train control systems.	Understand
CO4	Explain the planning, types and components of bridges, tunnels and airport facilities.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	Highway Planning, Alignment and Geometric Design	History of road development; highway planning and alignment factors; geometric design of cross-section elements; sight distances (SSD, OSD); horizontal and vertical alignment; highway drainage.	Approx. 14 hours	CO1	Module 1
2	Pavement Design, Materials and Maintenance	Types of pavements (flexible and rigid); pavement materials (soil, aggregate, bitumen); principles of pavement design (IRC methods); pavement evaluation and rehabilitation; road maintenance.	Approx. 16 hours	CO2	Module 2
3	Railway Engineering, Tracks and Signaling	Introduction to railways and gauges; permanent way components (rails, sleepers, ballast); geometric design of tracks; points and crossings; signaling and train control systems.	Approx. 16 hours	CO3	Module 3
4	Bridges, Tunnels and Airport Engineering	Bridge types and components; bridge site selection and maintenance; tunnel types, alignment and ventilation; airport planning, site selection and runway orientation; visual aids.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Highway Planning, Alignment and Geometric Design

History of road development; highway planning and alignment factors; geometric design of cross-section elements; sight distances (SSD, OSD); horizontal and vertical alignment; highway drainage.

CO1 · Approx. 14 hours

Pavement Design, Materials and Maintenance

Types of pavements (flexible and rigid); pavement materials (soil, aggregate, bitumen); principles of pavement design (IRC methods); pavement evaluation and rehabilitation; road maintenance.

CO2 · Approx. 16 hours

Railway Engineering, Tracks and Signaling

Introduction to railways and gauges; permanent way components (rails, sleepers, ballast); geometric design of tracks; points and crossings; signaling and train control systems.

CO3 · Approx. 16 hours

Bridges, Tunnels and Airport Engineering

Bridge types and components; bridge site selection and maintenance; tunnel types, alignment and ventilation; airport planning, site selection and runway orientation; visual aids.

CO4 · Approx. 14 hours

MODULE 1

Highway Planning, Alignment and Geometric Design

History of road development; highway planning and alignment factors; geometric design of cross-section elements; sight distances (SSD, OSD); horizontal and vertical alignment; highway drainage.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

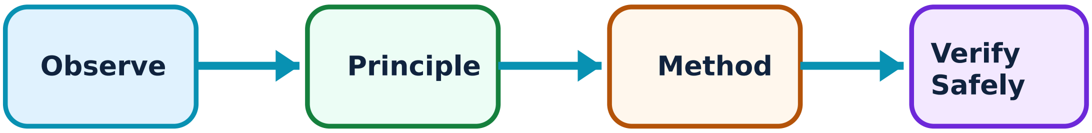
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Highway Planning, Alignment and Geometric Design: identify the system, state its principle, follow the correct method and verify the result safely.

1. Highway alignment and cross-section−

Highway alignment is the position of the centre line of a road. It must be selected for safety, economy and efficiency, considering terrain, soil and social factors. Cross-section elements like camber, kerbs and shoulders provide drainage, safety and structural support.

Study and application method

List the factors affecting highway alignment and sketch a typical rural highway cross-section showing all major elements.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the factors affecting highway alignment and sketch a typical rural highway cross-section showing all major elements.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Alignment decisions affect land acquisition and environment. Do not propose alignments without authorised surveying and environmental impact assessments.

2. Sight distances and curves−

Sight distance is the length of road visible to a driver. Stopping Sight Distance (SSD) and Overtaking Sight Distance (OSD) are critical for safety. Horizontal and vertical curves (summit, valley) are designed to provide smooth transitions and required sight distances.

Study and application method

Calculate the Stopping Sight Distance (SSD) for a given speed and gradient conceptually; explain the purpose of transition curves in horizontal alignment.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Geometric design must strictly follow current Indian Roads Congress (IRC) standards. Inadequate sight distance is a major cause of road accidents.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Pavement Design, Materials and Maintenance

Types of pavements (flexible and rigid); pavement materials (soil, aggregate, bitumen); principles of pavement design (IRC methods); pavement evaluation and rehabilitation; road maintenance.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

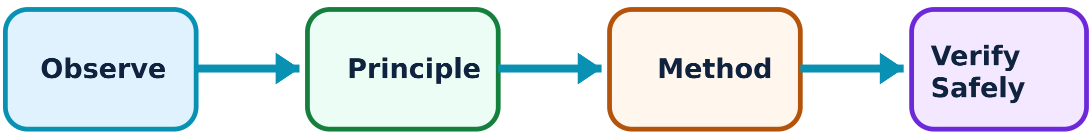
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Pavement Design, Materials and Maintenance: identify the system, state its principle, follow the correct method and verify the result safely.

1. Pavement types and materials−

Flexible pavements (bituminous) distribute loads through layers, while rigid pavements (concrete) use slab action. Materials like subgrade soil, aggregates and bitumen must meet specific quality standards for durability under traffic and climatic loads.

Study and application method

Compare the load-transfer mechanism of a flexible and a rigid pavement; describe three essential tests for road aggregates.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare the load-transfer mechanism of a flexible and a rigid pavement; describe three essential tests for road aggregates.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Pavement failure often starts with poor material quality or subgrade preparation. All road materials must be tested and approved as per authorised specifications.

2. Pavement design and maintenance−

Design involves determining layer thicknesses based on traffic volume and subgrade strength (CBR method). Maintenance (routine, periodic) and rehabilitation (overlays) are essential to preserve the road's serviceability and structural integrity over its life.

Study and application method

Outline the steps for designing a flexible pavement using the IRC method conceptually; identify three types of pavement distress and their causes.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Outline the steps for designing a flexible pavement using the IRC method conceptually; identify three types of pavement distress and their causes.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Pavement design requires accurate traffic forecasts and soil data. Do not treat a study-level design as an authorised construction specification.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Railway Engineering, Tracks and Signaling

Introduction to railways and gauges; permanent way components (rails, sleepers, ballast); geometric design of tracks; points and crossings; signaling and train control systems.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

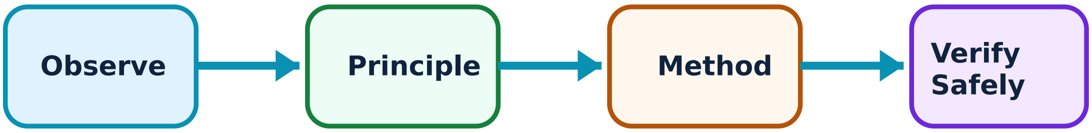
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Railway Engineering, Tracks and Signaling: identify the system, state its principle, follow the correct method and verify the result safely.

1. Permanent way and track components−

The permanent way consists of rails, sleepers, ballast and subgrade. Rails guide the wheels, sleepers maintain gauge and distribute loads, and ballast provides drainage and elasticity. Proper fastenings and joints are essential for track stability.

Study and application method

Sketch a cross-section of a permanent way on an embankment and label all components; describe the functions of ballast in a railway track.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a cross-section of a permanent way on an embankment and label all components; describe the functions of ballast in a railway track.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Track components are subject to extreme stresses. Any wear, creep or failure must be identified and corrected immediately by authorised railway personnel.

2. Track geometry and signaling−

Geometric design includes gauge, curves, gradients and superelevation (cant). Points and crossings allow trains to move between tracks. Signaling systems (mechanical, electrical) and interlocking ensure safe train movements and prevent collisions.

Study and application method

Define 'superelevation' and explain its importance on a railway curve; describe the working of a simple semaphore signal.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'superelevation' and explain its importance on a railway curve; describe the working of a simple semaphore signal.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Railway operations are highly regulated. Do not tamper with track components or signaling equipment, and ensure all work complies with authorised railway safety codes.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Bridges, Tunnels and Airport Engineering

Bridge types and components; bridge site selection and maintenance; tunnel types, alignment and ventilation; airport planning, site selection and runway orientation; visual aids.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

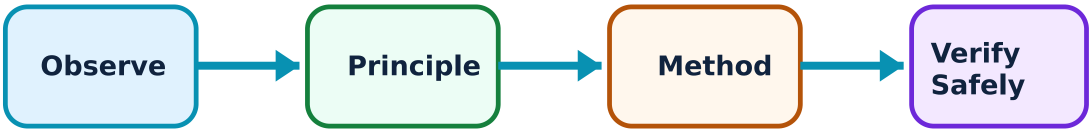
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Bridges, Tunnels and Airport Engineering: identify the system, state its principle, follow the correct method and verify the result safely.

1. Bridge and tunnel engineering−

Bridges cross obstacles like rivers or valleys; tunnels provide passage through hills or urban areas. Site selection, foundation type and structural form depend on geological and hydrological conditions. Ventilation and lighting are critical for tunnel safety.

Study and application method

Classify bridges based on structural form and describe the essential components of a bridge pier; list the factors affecting tunnel alignment.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Classify bridges based on structural form and describe the essential components of a bridge pier; list the factors affecting tunnel alignment.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Bridge and tunnel engineering involves high structural risks. Design and construction must be performed by specialised engineers following authorised bridge and tunnel codes.

2. Airport planning and runways−

Airport planning involves site selection based on topography, weather and traffic. Runway orientation is determined by wind data (wind rose). Geometric design of runways and taxiways, along with visual aids (markings, lighting), ensures safe aircraft operations.

Study and application method

Explain the purpose of a 'wind rose' diagram in runway orientation and list three visual aids used for night-time aircraft landings.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the purpose of a 'wind rose' diagram in runway orientation and list three visual aids used for night-time aircraft landings.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Aviation safety is paramount. Airport layouts and visual aids must strictly comply with national and international civil aviation regulations (e.g., DGCA/ICAO).

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Highway Planning, Alignment and Geometric Design

Use the concept flow to revise observation, principle, method and verification.

Module 2: Pavement Design, Materials and Maintenance

Use the concept flow to revise observation, principle, method and verification.

Module 3: Railway Engineering, Tracks and Signaling

Use the concept flow to revise observation, principle, method and verification.

Module 4: Bridges, Tunnels and Airport Engineering

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Sketch a typical highway cross-section for a two-lane road in a cutting.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the Stopping Sight Distance (SSD) for a vehicle on a level road conceptually.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare flexible and rigid pavements based on their construction and maintenance needs.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw a layout of a simple railway turnout and label the points and crossings.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Outline the factors considered in selecting a site for a new domestic airport.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Highway Planning, Alignment and Geometric Design

Explain Highway alignment and cross-section and state one correct method, application or precaution.−

Highway alignment is the position of the centre line of a road. It must be selected for safety, economy and efficiency, considering terrain, soil and social factors. Cross-section elements like camber, kerbs and shoulders provide drainage, safety and structural support.

Answer framework: List the factors affecting highway alignment and sketch a typical rural highway cross-section showing all major elements.

Important point: Alignment decisions affect land acquisition and environment. Do not propose alignments without authorised surveying and environmental impact assessments.

Explain Sight distances and curves and state one correct method, application or precaution.−

Sight distance is the length of road visible to a driver. Stopping Sight Distance (SSD) and Overtaking Sight Distance (OSD) are critical for safety. Horizontal and vertical curves (summit, valley) are designed to provide smooth transitions and required sight distances.

Answer framework: Calculate the Stopping Sight Distance (SSD) for a given speed and gradient conceptually; explain the purpose of transition curves in horizontal alignment.

Important point: Geometric design must strictly follow current Indian Roads Congress (IRC) standards. Inadequate sight distance is a major cause of road accidents.

Module 2 — Pavement Design, Materials and Maintenance

Explain Pavement types and materials and state one correct method, application or precaution.−

Flexible pavements (bituminous) distribute loads through layers, while rigid pavements (concrete) use slab action. Materials like subgrade soil, aggregates and bitumen must meet specific quality standards for durability under traffic and climatic loads.

Answer framework: Compare the load-transfer mechanism of a flexible and a rigid pavement; describe three essential tests for road aggregates.

Important point: Pavement failure often starts with poor material quality or subgrade preparation. All road materials must be tested and approved as per authorised specifications.

Explain Pavement design and maintenance and state one correct method, application

or precaution.−

Design involves determining layer thicknesses based on traffic volume and subgrade strength (CBR method). Maintenance (routine, periodic) and rehabilitation (overlays) are essential to preserve the road's serviceability and structural integrity over its life.

Answer framework: Outline the steps for designing a flexible pavement using the IRC method conceptually; identify three types of pavement distress and their causes.

Important point: Pavement design requires accurate traffic forecasts and soil data. Do not treat a study-level design as an authorised construction specification.

Module 3 — Railway Engineering, Tracks and Signaling

Explain Permanent way and track components and state one correct method, application or precaution.−

The permanent way consists of rails, sleepers, ballast and subgrade. Rails guide the wheels, sleepers maintain gauge and distribute loads, and ballast provides drainage and elasticity. Proper fastenings and joints are essential for track stability.

Answer framework: Sketch a cross-section of a permanent way on an embankment and label all components; describe the functions of ballast in a railway track.

Important point: Track components are subject to extreme stresses. Any wear, creep or failure must be identified and corrected immediately by authorised railway personnel.

Explain Track geometry and signaling and state one correct method, application or precaution.−

Geometric design includes gauge, curves, gradients and superelevation (cant). Points and crossings allow trains to move between tracks. Signaling systems (mechanical, electrical) and interlocking ensure safe train movements and prevent collisions.

Answer framework: Define 'superelevation' and explain its importance on a railway curve; describe the working of a simple semaphore signal.

Important point: Railway operations are highly regulated. Do not tamper with track components or signaling equipment, and ensure all work complies with authorised railway safety codes.

Module 4 — Bridges, Tunnels and Airport Engineering

Explain Bridge and tunnel engineering and state one correct method, application or precaution.−

Bridges cross obstacles like rivers or valleys; tunnels provide passage through hills or urban areas. Site selection, foundation type and structural form depend on geological and hydrological conditions. Ventilation and lighting are critical for tunnel safety.

Answer framework: Classify bridges based on structural form and describe the essential components of a bridge pier; list the factors affecting tunnel alignment.

Important point: Bridge and tunnel engineering involves high structural risks. Design and construction must be performed by specialised engineers following authorised bridge and tunnel codes.

Explain Airport planning and runways and state one correct method, application or precaution.−

Airport planning involves site selection based on topography, weather and traffic. Runway orientation is determined by wind data (wind rose). Geometric design of runways and taxiways, along with visual aids (markings, lighting), ensures safe aircraft operations.

Answer framework: Explain the purpose of a 'wind rose' diagram in runway orientation and list three visual aids used for night-time aircraft landings.

Important point: Aviation safety is paramount. Airport layouts and visual aids must strictly comply with national and international civil aviation regulations (e.g., DGCA/ICAO).

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Highway Planning, Alignment and Geometric Design.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Pavement Design, Materials and Maintenance.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Railway Engineering, Tracks and Signaling.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Bridges, Tunnels and Airport Engineering.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: History of road development; highway planning and alignment factors; geometric design of cross-section elements; sight distances (SSD, OSD); horizontal and vertical alignment; highway drainage..
2. Develop a complete answer or practical plan for: Types of pavements (flexible and rigid); pavement materials (soil, aggregate, bitumen); principles of pavement design (IRC methods); pavement evaluation and rehabilitation; road maintenance..
3. Develop a complete answer or practical plan for: Introduction to railways and gauges; permanent way components (rails, sleepers, ballast); geometric design of tracks; points and crossings; signaling and train control systems..
4. Develop a complete answer or practical plan for: Bridge types and components; bridge site selection and maintenance; tunnel types, alignment and ventilation; airport planning, site selection and runway orientation; visual aids..

LAST-MILE REVISION

Quick Revision

Module 1: Highway Planning, Alignment and Geometric Design

History of road development; highway planning and alignment factors; geometric design of cross-section elements; sight distances (SSD, OSD); horizontal and vertical alignment; highway drainage.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Pavement Design, Materials and Maintenance

Types of pavements (flexible and rigid); pavement materials (soil, aggregate, bitumen); principles of pavement design (IRC methods); pavement evaluation and rehabilitation; road maintenance.

- Match each topic to CO2.

- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Railway Engineering, Tracks and Signaling

Introduction to railways and gauges; permanent way components (rails, sleepers, ballast); geometric design of tracks; points and crossings; signaling and train control systems.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Bridges, Tunnels and Airport Engineering

Bridge types and components; bridge site selection and maintenance; tunnel types, alignment and ventilation; airport planning, site selection and runway orientation; visual aids.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Highway alignment and cross-section	Highway alignment is the position of the centre line of a road.
Sight distances and curves	Sight distance is the length of road visible to a driver.
Pavement types and materials	Flexible pavements (bituminous) distribute loads through layers, while rigid pavements (concrete) use slab action.
Pavement design and maintenance	Design involves determining layer thicknesses based on traffic volume and subgrade strength (CBR method).
Permanent way and track components	The permanent way consists of rails, sleepers, ballast and subgrade.
Track geometry and signaling	Geometric design includes gauge, curves, gradients and superelevation (cant).
Bridge and tunnel engineering	Bridges cross obstacles like rivers or valleys; tunnels provide passage through hills or urban areas.
Airport planning and runways	Airport planning involves site selection based on topography, weather and traffic.

LEARNING RESOURCES

References

Recommended transportation-engineering references

1. S. K. Khanna and C. E. G. Justo, Highway Engineering.
2. S. C. Saxena and S. P. Arora, A Text Book of Railway Engineering.
3. S. P. Bindra, A Course in Bridge Engineering.
4. Current IRC codes and Railway/Aviation manuals as applicable.

Approved public transportation-engineering sources

- https://nptel.ac.in/courses/105105107
- https://nptel.ac.in/courses/105107123

These sources provide the verified study basis while official Course 6011T detail is unavailable; they do not replace official transport planning data, authorised design codes, traffic regulations or project-specific engineering studies.